

Intro to Data Science

Lab 1

Descriptive Analytics for Business

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1 Key Metrics Analysis

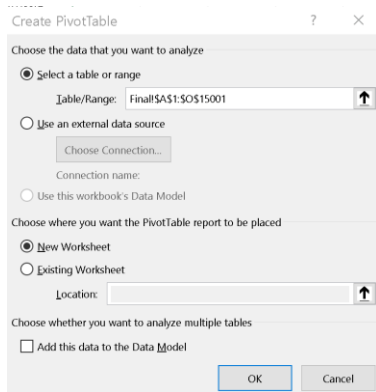
Key Metrics or Key Performance Indicators (KPIs) are the main method that we use to measure the performance of a company, division, department or team. Key Metrics provide an overall understanding of how the business is performing. Key Metrics usually include the following types of questions:

- a) What is the Total Sales?
- b) What is the Total Profit?
- c) What is the Profit Ratio?
- d) How many transactions were there processed?
- e) What was the average Sales amount?
- f) What was the highest Sales amount?
- g) What was the lowest Sales amount?

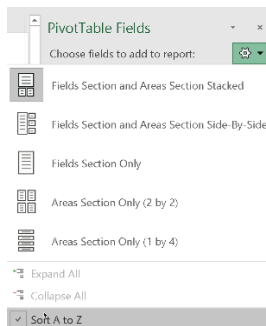
File to use: SalesData-v1.xlsx

Create a Pivot table from the current range of data.
Make sure you have selected a cell in the range.
Insert -> Pivot table

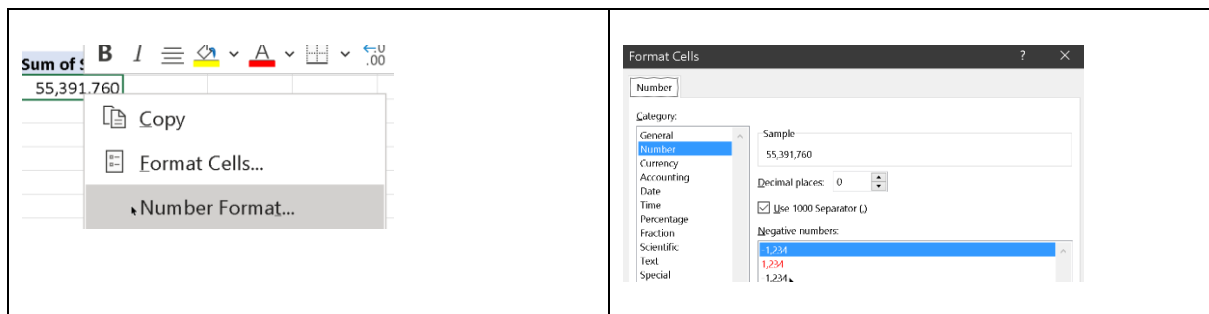
The Create Pivot table should automatically detect the complete range of cells with valid values. Make sure you create the Pivot table in a new worksheet.



You can change the order of the Pivot Table fields as they appear in the main list: its sometimes useful to sort them in alphabetical order if you have many fields.



Select the Sales field to get the Sum of Sales, then perform a Number Format to make it easier to view.



Repeat this with the Profit field to get the Sum of Profit, with a similar Number Format as well.

Sum of Sales	Sum of Profit
55,391,760	35,589,375

We can now create a Calculated Field to obtain the Profit Ratio.

With at least once cell selected in the pivot table, select Pivot Table Analyze from the top main menu, select Fields, Items and Sets

Next insert a Calculated Field to obtain the Profit Ratio, and format the result accordingly

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The screenshot shows two overlapping dialog boxes in Power BI Desktop. The 'Insert Calculated Field' dialog is in the foreground, with 'Name' set to 'Profit Ratio' and 'Formula' set to '= Profit / Sales'. The 'Fields' list on the left includes 'Order Date', 'Year', 'Order Qty', 'Cost of Sales', 'Sales', 'Profit', 'Channel', and 'Product Name'. The 'Format Cells' dialog is open in the background, showing the 'Number' category with 'Sample' set to '64.3%' and 'Decimal places' set to '1'.

Sum of Sales	Sum of Profit	Sum of Profit Ratio
55,391,760	35,589,375	64.3%

To count the number of transactions, we would typically use a unique Sales ID, which we don't have here in this dataset. We can initially use the Sales Field to start with (drag and drop into the Values area to obtain a new Sum of Sales 2).

Then select this cell and from the context menu, change it to Summarize Values by Count to get the actual number of transactions (which we can also double confirm from the source dataset)

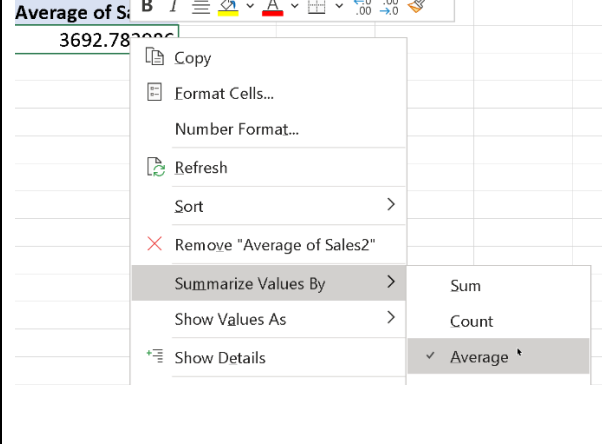
The screenshot shows a context menu for the 'Count of Sales' field. The menu options include 'Copy', 'Format Cells...', 'Number Format...', 'Refresh', 'Sort', 'Remove "Count of Sales2"', 'Summarize Values By' (with 'Sum' selected), and 'Show Values As' (with 'Count' selected). To the right, a table displays the results of these actions.

Sum of Sales	Sum of Profit	Sum of Profit Ratio	Count of Sales2
55,391,760	35,589,375	64.3%	15000

To get the average Sales, use the Sales field again in the same manner (drag and drop into the Values area to obtain a new Sum of Sales 2).

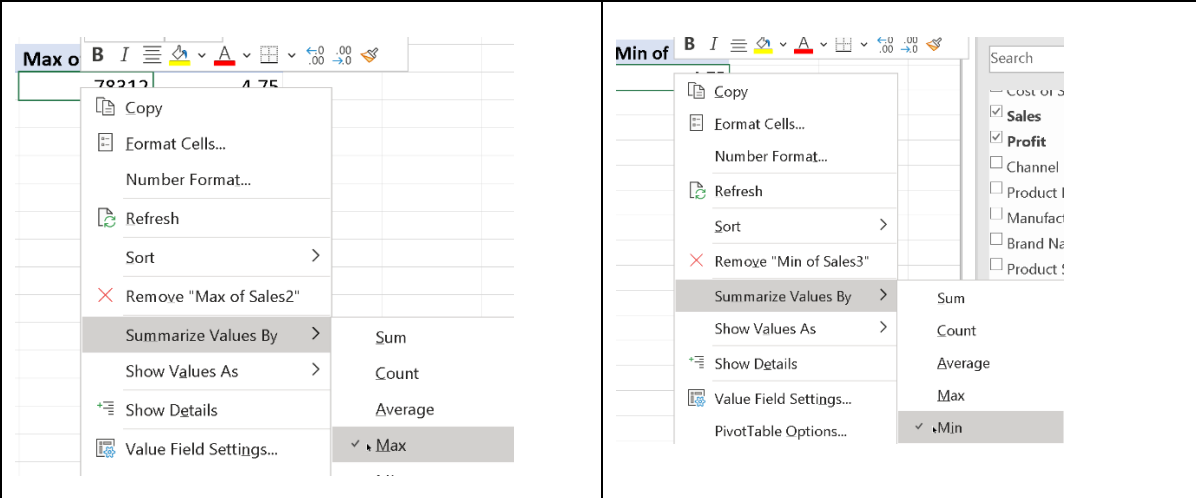
Then select this cell and from the context menu, change it to Summarize Values by Average to get the average Sales value, then format it accordingly.

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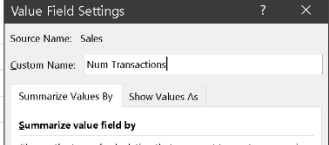
Count of Sales2	Average of Sales2
15000	3,693

Add two more Sales Fields into the Values area in a similar manner as previously, and then perform a Summarize Values by Min and Max for these two new fields and format them appropriately.



Now that we have all our numbers, you can double click on any (or all) of the field headers to change the field names to something more appropriate if you wish.

Profit Ratio	Count of Sales2	Average of Sales2
64.3%	15000	3,693



We now have our key metrics listed out in a row:

Sum of Sales	Sum of Profit	Sum of Profit Ratio	Num Transactions	Average of Sales	Highest Sales	Lowest Sales
55,391,760	35,589,375	64.3%	15000	3,693	78312	5

You can transpose the rows and columns for a better layout:

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Drag fields between areas below: Filters Columns Rows Σ Values Σ Values Sum of Sales Sum of Profit Sum of Profit Ratio	<table> <thead> <tr> <th colspan="2">Values</th></tr> </thead> <tbody> <tr> <td>Total Sales</td><td>55,391,760</td></tr> <tr> <td>Total Profit</td><td>35,589,375</td></tr> <tr> <td>Actual Profit Ratio</td><td>64.3%</td></tr> <tr> <td>Num Transactions</td><td>15000</td></tr> <tr> <td>Average of Sales</td><td>3,693</td></tr> <tr> <td>Highest Sales</td><td>78312</td></tr> <tr> <td>Lowest Sales</td><td>5</td></tr> </tbody> </table>	Values		Total Sales	55,391,760	Total Profit	35,589,375	Actual Profit Ratio	64.3%	Num Transactions	15000	Average of Sales	3,693	Highest Sales	78312	Lowest Sales	5
Values																	
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Total Profit	35,589,375																
Actual Profit Ratio	64.3%																
Num Transactions	15000																
Average of Sales	3,693																
Highest Sales	78312																
Lowest Sales	5																

To zoom in and get more details on the metrics based on a certain category (for e.g. country), we can filter the table to decide which subset of records to apply the various aggregation operations we have performed so far on.

For this purpose, we can use a slicer. Go to Pivot Table Analyze -> Insert Slicer, and select Country. You can then select particular countries from the slicer drop down to get the metric values only for sales records related to that particular country.

viewViewHelpAnalyzeDesign

tion

Insert Slicer

Insert Timeline

Filter

Refresh

Insert Slicers

☐ Order Date

☐ Year

☐ Order Qty

☐ Cost of Sales

☐ Sales

☐ Profit

☐ Channel

☐ Product Name

☐ Manufacturer

☐ Brand Name

☐ Product Sub Category

☐ Product Category

☐ Region

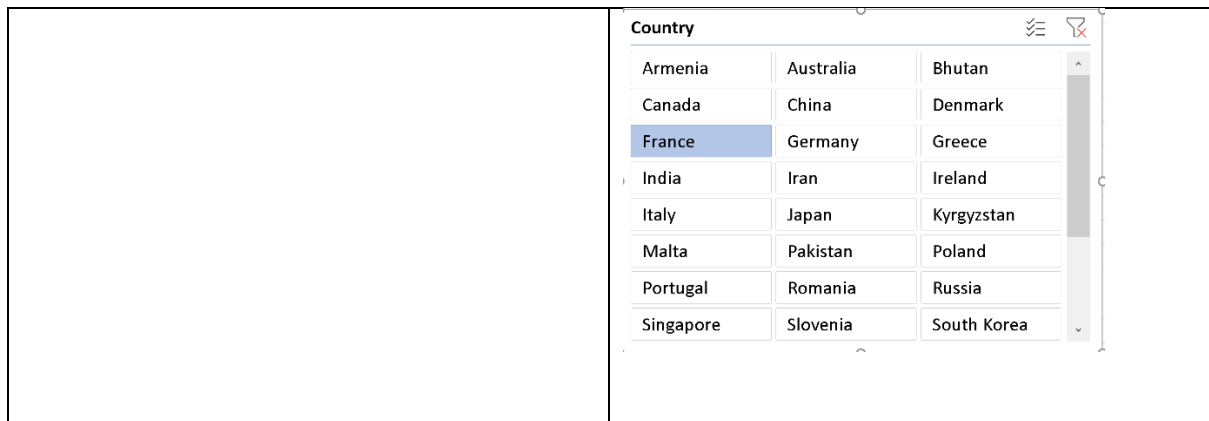
☐ City

☒ Country

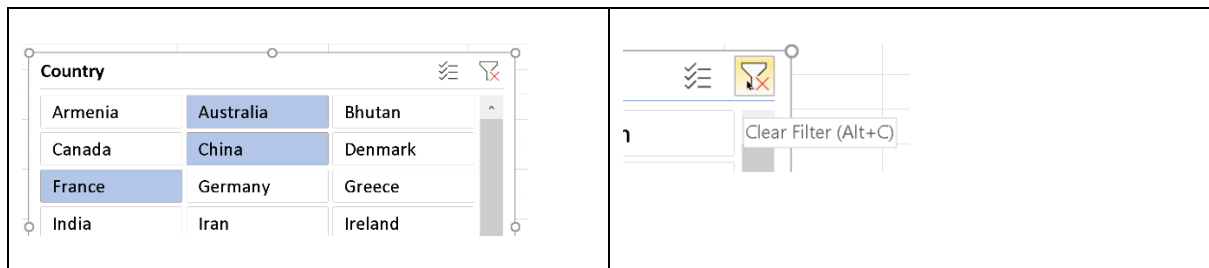
Values		Country
Total Sales	1,148,316	Armenia
Total Profit	694,190	Australia
Actual Profit Ratio	60.5%	Bhutan
Num Transactions	359	Canada
Average of Sales	3,199	China
Highest Sales	45900	Denmark
Lowest Sales	37	France
		Germany

With the slicer selected and then selecting the Slicer option in the main menu, you can fine tune the slicer by adjusting the number of columns, as well as their height and width (you can just click and drag on the slicer itself to manually change this) to make them easier to work with:

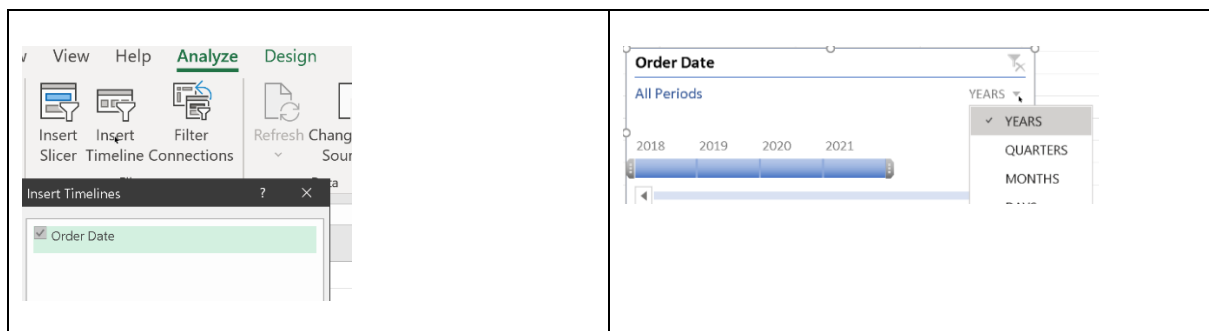
Slicer Tools SalesData v1.xlsx Options Search Columns: 3 Height: 0.67 cm Width: 2.88 cm Height: 7.01 cm Width: 9.87 cm	
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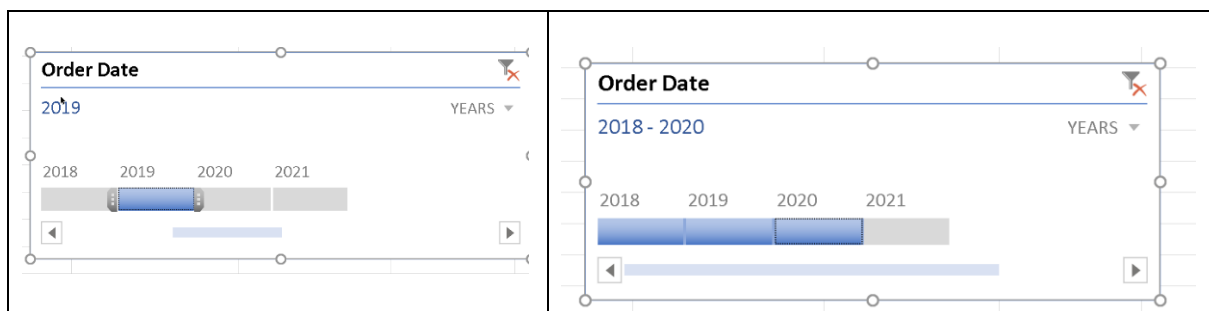
You can use the Ctrl+Left mouse click to select multiple countries at the same time, so the aggregation operations will apply to all these countries. You can then select the clear Filter button to clear the filter and make the aggregation apply to the entire dataset.



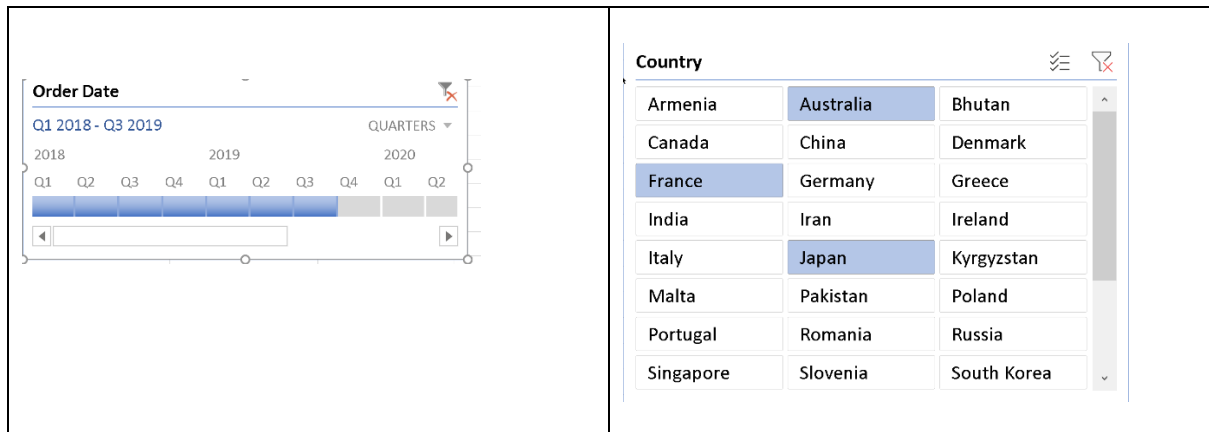
You can now also insert a Timeline and decide which particular period (Years, Quarters, Months) to examine.



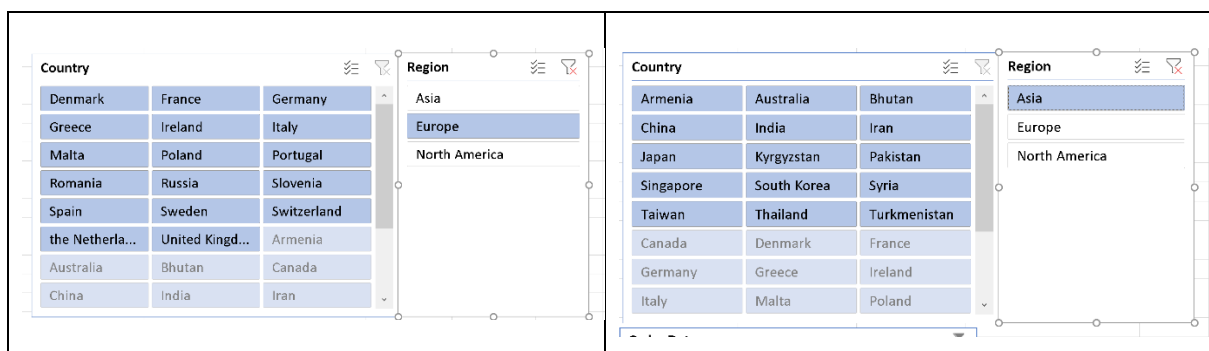
You can select a particular period or a range of periods (Shift-Click):



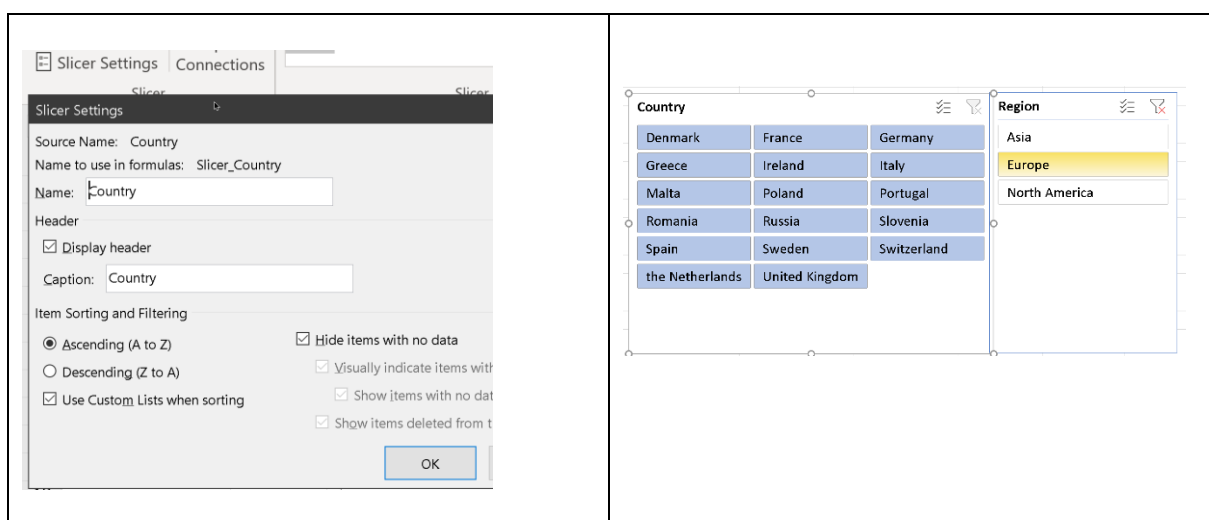
You can also select items from both filters to combine their filtering action together, for e.g. to perform aggregation operations on a specific country (or group of countries) over a specific period of time.



You can introduce another slicer for Region, and notice that selection of items in either slicer for Country or Region will influence the other: based on which countries are included in which region. To see this properly in effect, you will have to clear the filter in either one or both of the slicers.



You can also set the Slicer Settings to only clearly show the countries within a particular region, and not just grey out the countries that are not in that region (to make the analysis even more clearer):



1.1 Practical Exercise for Key Metrics Analysis

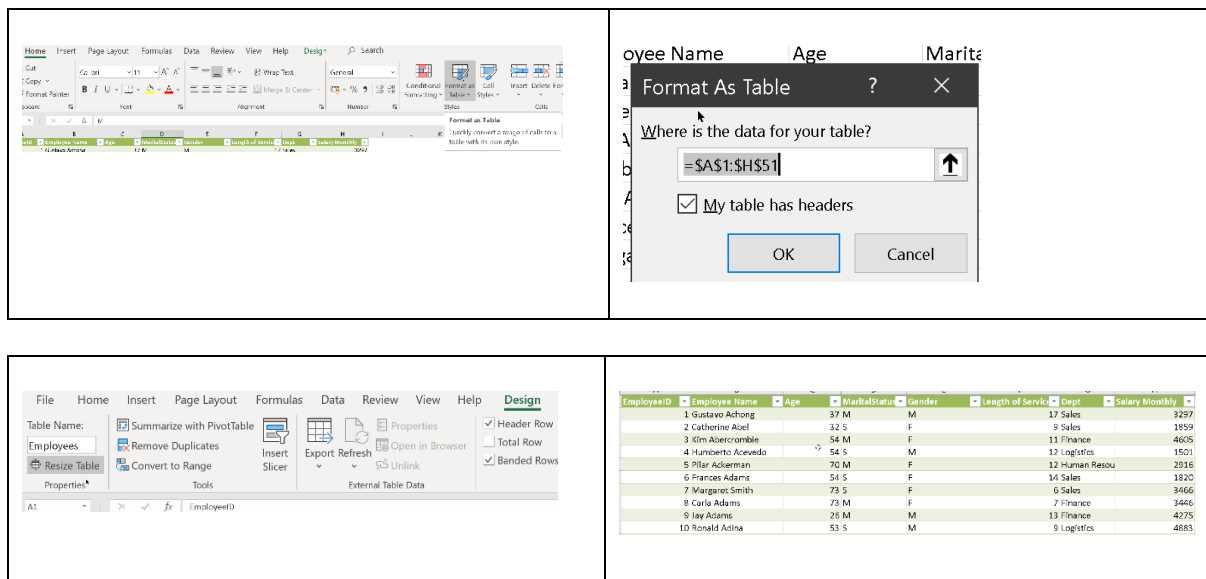
File to use: Employee-v1.xlsx

Analytical activities to perform:

1. How many employees are in John's company?
2. What is the average age of the employees?
3. What is the highest age at the company?
4. What is the lowest age at the company?
5. What is the average Length of Service?
6. What is the longest Length of Service?

Drill down further into this information based on the following categories: Dept and Gender

You can initially format the range of cells of the original data set as a table to make it easier to work with, and to create a Pivot Table. To do this, select a cell in the data range, go to Home -> Format as Table, format it and give it an appropriate name.

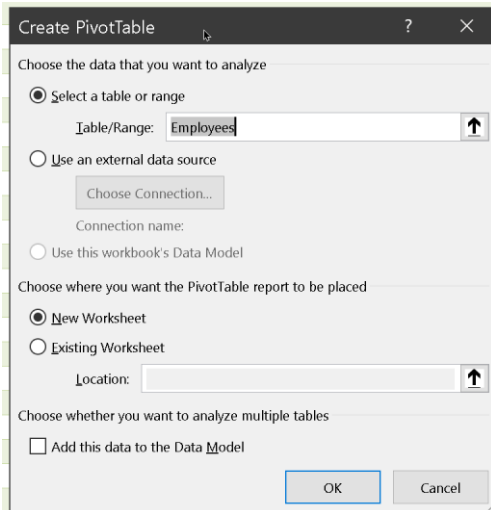


The screenshot shows the Excel interface with the 'Format as Table' dialog box open. The dialog box asks 'Where is the data for your table?' and shows the range '\$A\$1:\$H\$51'. The checkbox 'My table has headers' is checked. The 'OK' button is highlighted.

Below the dialog box, the resulting table is shown. The table has 10 columns: EmployeeID, Employee Name, Age, MaritalStatus, Gender, Length of Service, Dept, and Salary Monthly. The data is as follows:

EmployeeID	Employee Name	Age	MaritalStatus	Gender	Length of Service	Dept	Salary Monthly
1	Gustavo Achong	37	M	M	17	Sales	3297
2	Catherine Acel	32	S	F	9	Sales	1859
3	Kim Abercrombie	54	M	F	11	Finance	4605
4	Humberto Acevedo	54	S	M	12	Logistics	1501
5	Pilar Ackerman	70	M	F	12	Human Resou	2916
6	Frances Adams	54	S	F	14	Sales	1620
7	Margaret Smith	73	S	F	6	Sales	3466
8	Carla Adams	73	M	F	7	Finance	3446
9	Jay Adams	26	M	M	13	Finance	4275
10	Ronald Adina	53	S	M	9	Logistics	4883

Once done, you can generate a Pivot Table in the usual way, but this time referencing the Table.



2 Comparison Analysis

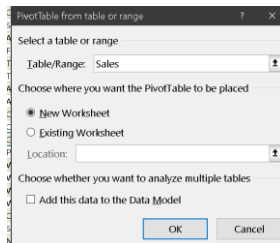
Comparison analysis is probably one of the most popular forms of analysis you're going to do. The focus is finding the magnitude of difference between values for different fields. Visualizing your data set makes it much easier to appreciate this as opposed to directly examining the values themselves.

Sample analysis

- a) compare the sales and profit by year
- b) compare the sales by manufacturer
- c) compare the profit by product categories
- d) compare the sales by product category and by channel
- e) compare the profit by product subcategory and review that by different countries.
- f) compare the sales and profit by the countries, and review that by different product sub categories

File to use: SalesData-v2.xlsx

Start off again by generating a Pivot Table in the usual manner that references the Sales table.



In this new Pivot table, obtain the Sum of Sales and Profit by Years (Rows) and format the cells with an appropriate numeric format (select all the cells -> Format Cells)

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Drag fields between areas below:			
Filters	Columns		
	Σ Values		
Rows	Σ Values		
Year	Sum of Sales		
	Sum of Profit		

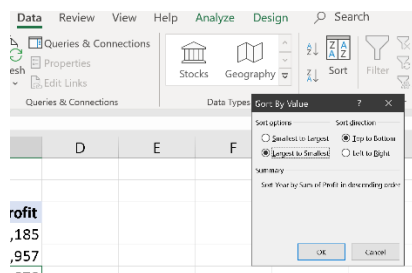
Row Labels	Sum of Sales	Sum of Profit
2018	18,919,151	12,057,185
2019	17,741,637	11,378,957
2020	17,284,124	11,237,878
2021	1,446,849	915,355
Grand Total	55,391,760	35,589,375

We can also sort on the Sum of Sales or Profit (either Smallest to Largest or vice versa)

Row Labels	Sum of Sales	Sum of Profit
2018	18,919,151	12,057,185
2019	17,741,637	11,378,957
2020	17,284,124	11,237,878
2021	1,446,849	915,355
Grand Total	55,391,760	35,589,375

Sort	Sort Smallest to Largest
Remove "Sum of Sales"	Sort Largest to Smallest
Summarize Values By	More Sort Options...

The sorting can also be done via the Sort option in the Data tab



Sort in descending order for Sum of Sales, and visualize this via a Pivot Chart (choose Clustered Column)

Analyze Design Search

Filter injections Refresh Change Data Source Clear Select Move PivotTable Fields, Items, & Sets OLAP Relationships Calculations PivotChart

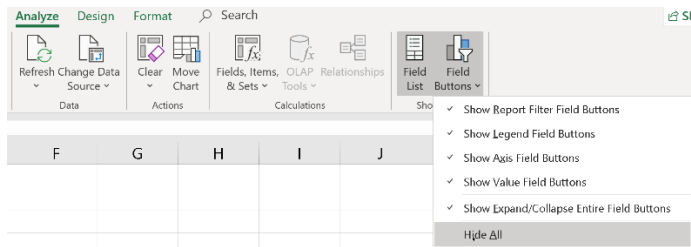
Insert Chart

All Charts

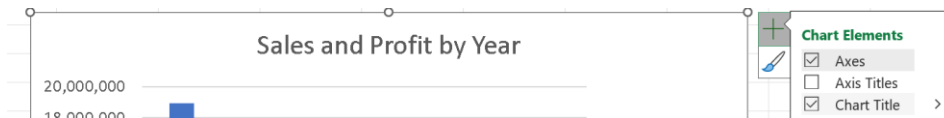
Recent Templates Column

Clustered Column

You can format what fields to be shown in the chart, including hiding everything:



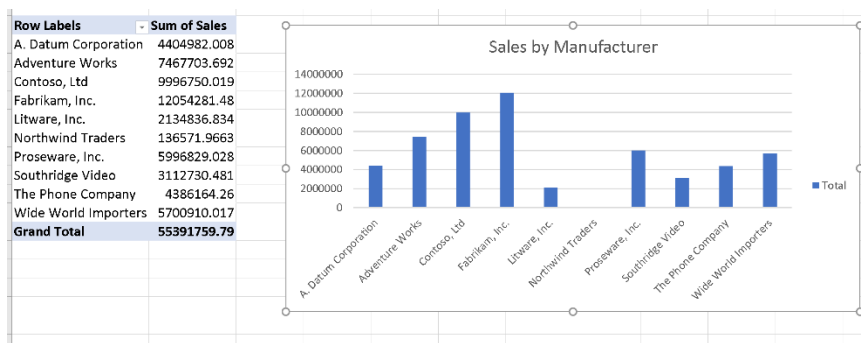
You can also change the Chart Elements appropriately, for e.g. add in Title and moving the legend to the Top.



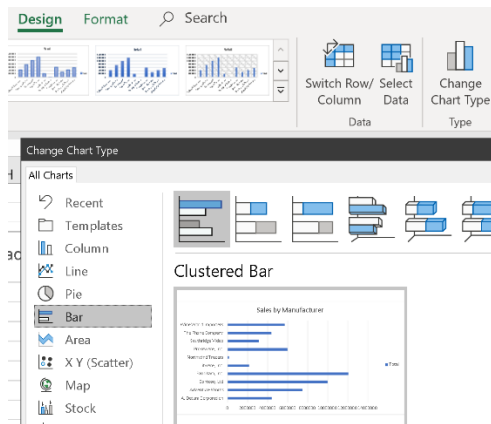
We can now create a new Pivot Table and repeat the earlier process to get the Sum of Sales by Manufacturer (Rows) and format the cells with an appropriate numeric format.

Row Labels	Sum of Sales
A. Datum Corporation	4404982.008
Adventure Works	7467703.692
Contoso, Ltd	9996750.019
Fabrikam, Inc.	12054281.48
Litware, Inc.	2134836.834
Northwind Traders	136571.9663
Proseware, Inc.	5996829.028
Southridge Video	3112730.481
The Phone Company	4386164.26
Wide World Importers	5700910.017
Grand Total	55391759.79

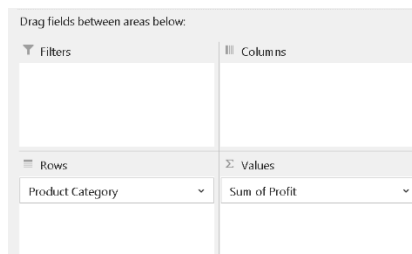
We can now insert another Clustered Column chart, and format it in the same way as we did previously.



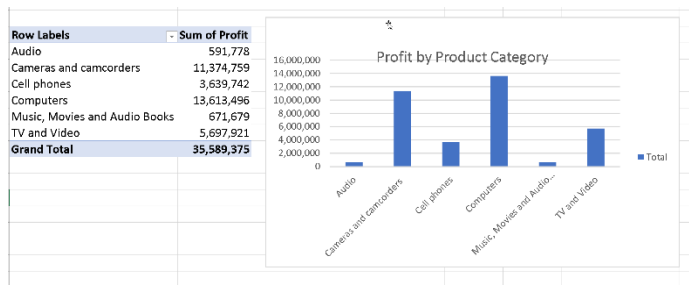
We can also opt to change the Chart Type to another type, for example Clustered Bar.



For the next analysis, we can again create a new Pivot Table and repeat the earlier process to get the Sum of Profit by Product Category (Rows) and format the cells with an appropriate numeric format



We can then generate the Pivot Chart (either Clustered Column or Clustered Bar).

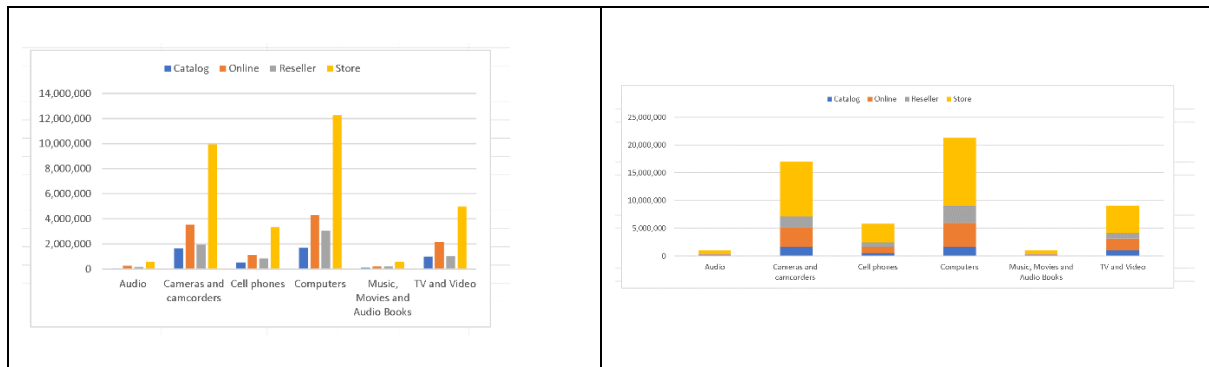


For the next analysis, we can again create a new pivot table and repeat the earlier process to get the Sum of Sales by Product Category (Rows) and Channel (Columns) and format the cells with an appropriate numeric format.

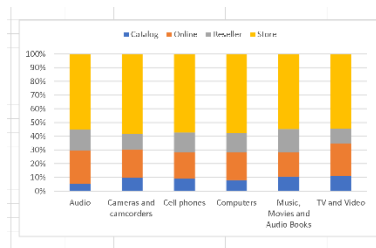
Drag fields between areas below:					
Filters		Columns			
		Channel			
Rows		Values			
Product Categ...		Sum of Sales			

Sum of Sales					
Column Labels					
Row Labels	Catalog	Online	Reseller	Store	Grand Total
Audio		53,274	248,765	159,539	567,803
Cameras and camcorders		1,641,341	3,514,850	1,933,395	9,959,345
Cell phones		527,490	1,114,245	845,416	3,354,004
Computers		1,682,876	4,290,928	3,045,515	12,287,652
Music, Movies and Audio Books		111,079	185,645	181,045	579,712
TV and Video		984,231	2,145,775	1,005,703	4,972,131
Grand Total		5,000,291	11,500,207	7,170,614	31,720,647

Then we could use either or both a Clustered Column chart and / or Stacked Column chart to view the differences.



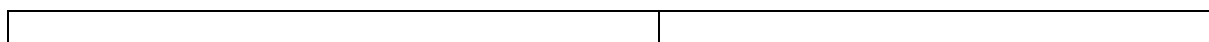
We could also use a 100% stacked column as well, which is useful for seeing how each of the different channels contribute for all products, regardless of the product total - this is useful for products whose sales totals are very small compared to other products and therefore will not come out clearly in the previous graphs.

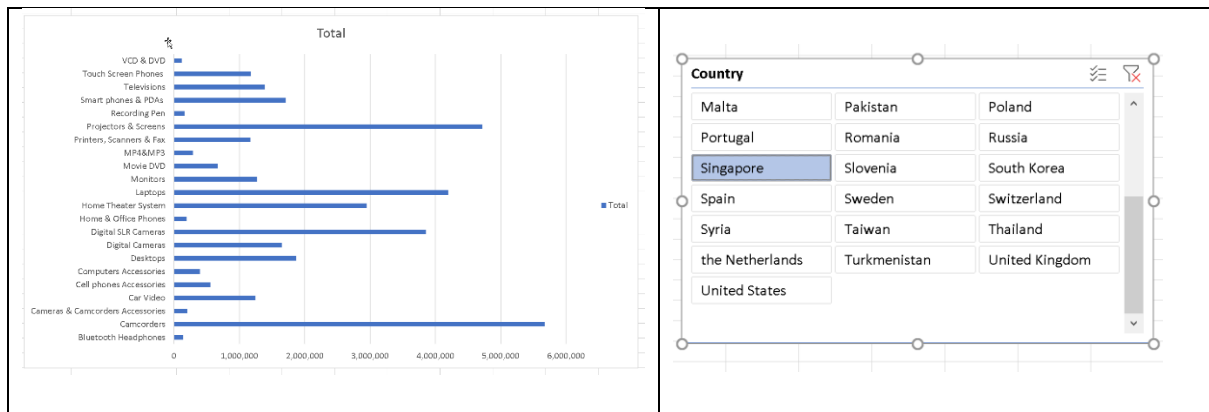


For the next analysis, we can again create a new pivot table and repeat the earlier process to get the Sum of Profit by Product Sub Category (Rows) and format the cells with an appropriate numeric format.

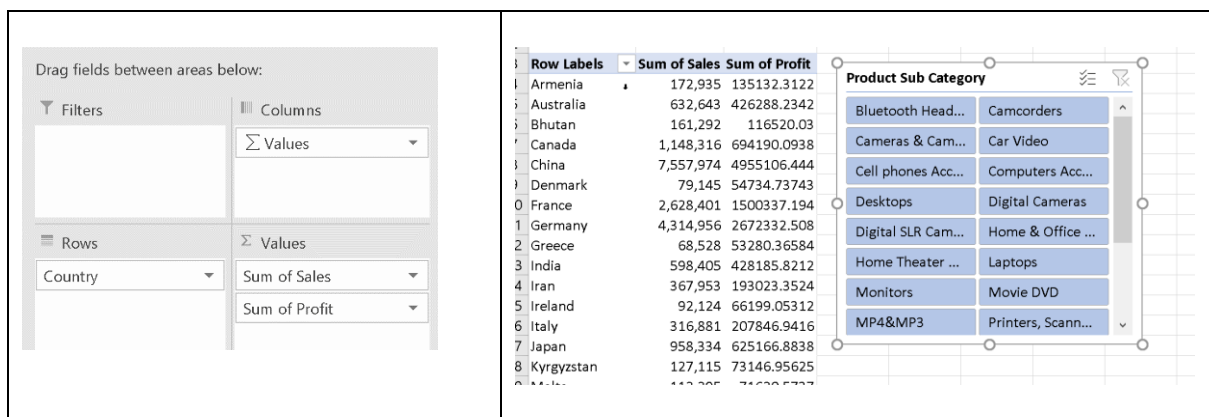
Axis (Categories) Product Sub Category Σ Values Sum of Profit	<table> <thead> <tr> <th>Row Labels</th><th>Sum of Profit</th></tr> </thead> <tbody> <tr><td>Bluetooth Headphones</td><td>142,887</td></tr> <tr><td>Camcorders</td><td>5,670,098</td></tr> <tr><td>Cameras & Camcorders Accessories</td><td>202,945</td></tr> <tr><td>Car Video</td><td>1,243,947</td></tr> <tr><td>Cell phones Accessories</td><td>558,835</td></tr> <tr><td>Computers Accessories</td><td>395,441</td></tr> <tr><td>Desktops</td><td>1,870,700</td></tr> <tr><td>Digital Cameras</td><td>1,647,076</td></tr> <tr><td>Digital SLR Cameras</td><td>3,854,640</td></tr> <tr><td>Home & Office Phones</td><td>198,426</td></tr> <tr><td>Home Theater System</td><td>2,946,634</td></tr> <tr><td>Laptops</td><td>4,193,350</td></tr> <tr><td>Monitors</td><td>1,268,828</td></tr> <tr><td>Movie DVD</td><td>671,679</td></tr> <tr><td>MP4&MP3</td><td>288,293</td></tr> <tr><td>Printers, Scanners & Fax</td><td>1,167,342</td></tr> <tr><td>Projectors & Screens</td><td>4,717,835</td></tr> <tr><td>Recording Pen</td><td>160,597</td></tr> <tr><td>Smart phones & PDAs</td><td>1,708,954</td></tr> <tr><td>Televisions</td><td>1,388,547</td></tr> <tr><td>Touch Screen Phones</td><td>1,173,528</td></tr> <tr><td>VCD & DVD</td><td>118,793</td></tr> <tr><td>Grand Total</td><td>35,589,375</td></tr> </tbody> </table>	Row Labels	Sum of Profit	Bluetooth Headphones	142,887	Camcorders	5,670,098	Cameras & Camcorders Accessories	202,945	Car Video	1,243,947	Cell phones Accessories	558,835	Computers Accessories	395,441	Desktops	1,870,700	Digital Cameras	1,647,076	Digital SLR Cameras	3,854,640	Home & Office Phones	198,426	Home Theater System	2,946,634	Laptops	4,193,350	Monitors	1,268,828	Movie DVD	671,679	MP4&MP3	288,293	Printers, Scanners & Fax	1,167,342	Projectors & Screens	4,717,835	Recording Pen	160,597	Smart phones & PDAs	1,708,954	Televisions	1,388,547	Touch Screen Phones	1,173,528	VCD & DVD	118,793	Grand Total	35,589,375
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Then we generate a standard Clustered Bar Chart as we have done before, and now we can also choose to use a slicer based on the Country field for the pivot table which will also dynamically affect the content of the chart which is linked to the same table.





For the next analysis, we can again create a new pivot table and repeat this earlier process to get the Sum of Sales and Profit by Country (Rows) and format the cells with an appropriate numeric format. Then insert a new Slicer based on Product Subcategory



2.1 Practical exercise for comparison Analysis

File to use: File to use: SalesData-v3.xlsx

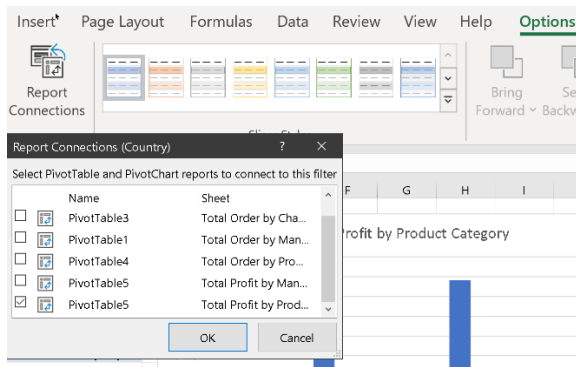
Analytical activities to perform:

1. What is the Total Order Quantity by Manufacturer?
2. What is the average Profit by Brand
3. Create a column graph displaying Total Order Quantity sold by Channel - review by Country
4. Create a bar graph displaying the Total Order quantity sold by Product Category - review by Region

Create two pivot graphs as per the instructions below. Filter both graphs by the Country and Year fields using a slicer.

5. Create a column graph displaying Total Profit by Product Category
6. Create a bar graph displaying Total Profit by Manufacturer

Note: While creating slicers, you can link a single slicer to multiple pivot tables on the same worksheet (if you decide to have more than one), using Report Connections in the Slicer Options.



3 Trend Analysis

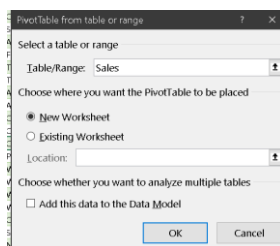
Trend analysis is all about understanding how data changes over time: is it moving up, is it moving down, is it staying stable or is it volatile. This can be accomplished mainly via trendlines. We also want to study these changes over different time frames: for e.g. changes over years, quarters, months, days. Again, this is best understood with some form of visualization.

Activities to be undertaken:

- Understand the trend for sales by year and month
- Understand the trend for sales by year and month by different product categories
- Develop a seasonality graph displaying the sales by month

File to use: File to use: SalesData-v4.xlsx

Start off again by generating a Pivot Table in the usual manner that references the Sales table.



Adding Order Date to the Rows area results in Excel creating a hierarchy grouping involving Years and Quarters or Years and Months.

NOTE: You may get a slightly different result from that shown below depending on the Grouping option that you had created for this column from a previous exercise – that is ok because we will ungroup and regroup again with our custom grouping later.

--	--

Rows Years (Order Date) Months (Order Date) Order Date	Row Labels 2018 Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec 2019
--	---

You can ungroup the hierarchy first before creating your own custom grouping based only on years.

Row Labels 2018 2019 2020 2021 Grand Total	Row Labels 2/2/2018 3/2/2018 4/2/2018 5/2/2018 6/2/2018 7/2/2018 8/2/2018 9/2/2018 10/2/2018 11/2/2018 12/2/2018 13/2/2018 14/2/2018 15/2/2018 16/2/2018 17/2/2018 18/2/2018 19/2/2018
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We can then continue to get Sum of Sales for each year

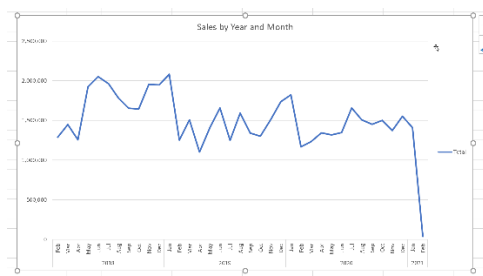
Rows Order Date	Σ Values Sum of Sales	Row Labels Sum of Sales 2018 2019 2020 2021 Grand Total
--	--	---

We can also alternatively change to another custom grouping (years / months)

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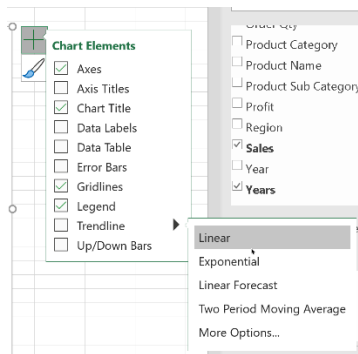
Row Labels	Sum of Sales
2018	18,919,151
Feb	1,289,552
Mar	1,448,250
Apr	1,256,708
May	1,923,673
Jun	2,051,973
Jul	1,965,917
Aug	1,781,769
Sep	1,654,111
Oct	1,643,812
Nov	1,955,526
Dec	1,947,858
2019	17,741,637
Jan	2,081,908

We will insert a Pivot Chart in the form of a Line Chart.



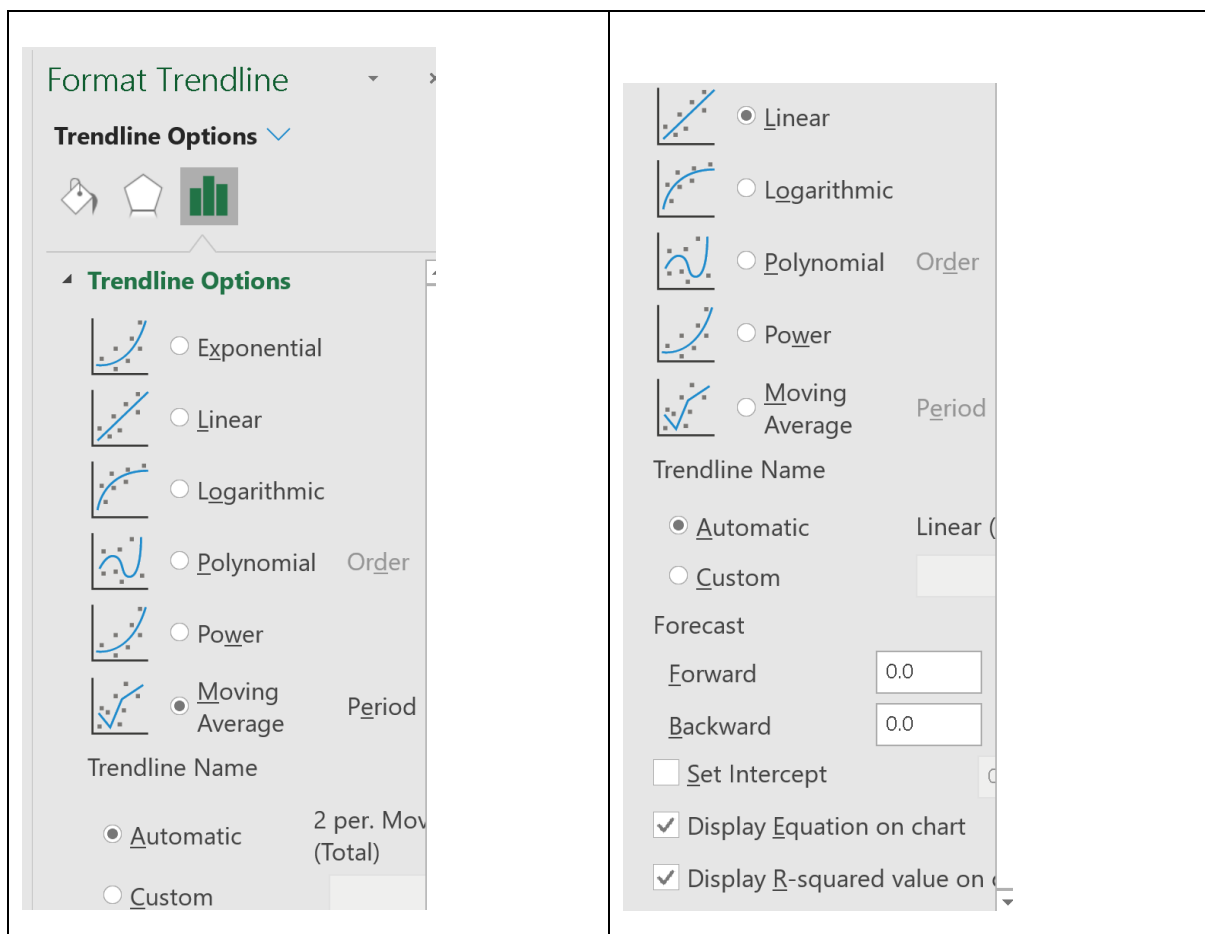
We can also filter out the first 2 months of 2021 to avoid issues with insufficient data at that point which will affect the trend analysis. Make sure the Row Labels are only showing the years (and not the months as well) and click on the drop down arrow.

With a more proper graph, you can examine the various trend line options possible:



The first option is the default option which is linear. The rest (exponential, linear forecast) are quite similar, but not exactly the same. They are both using different algorithms to calculate what the trend line is. The two period moving average is basically taking two periods and it averages that value between the two periods. It tends to smooth the lines. Its not a trend line as such but tracking the shape of the sales lines.

Selecting More Options leads to the Format Trend line menu, which provides a variety of options such as polynomial, exponential, etc.



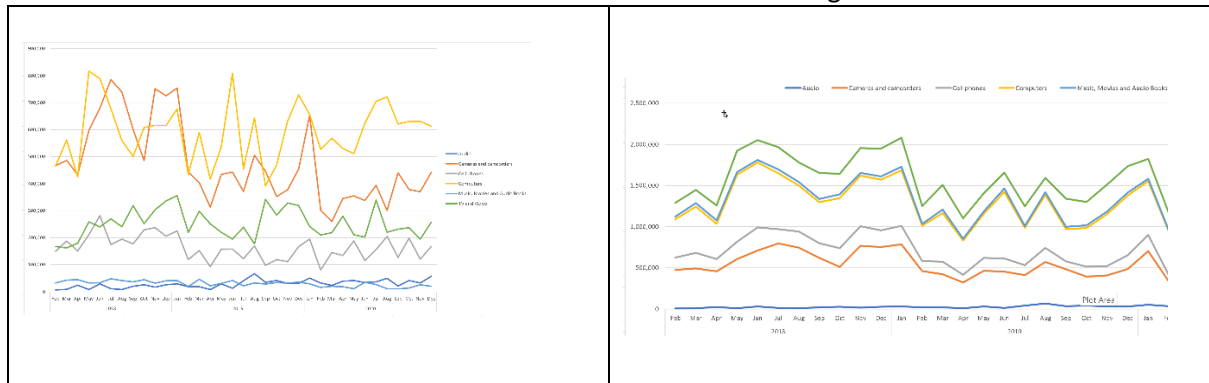
For a linear equation, you can display the trend line equation on the chart and calculate the R-squared, which is a measure of how accurate the predictions would be from the trend line equation.

The next step is to refine this further to understand the trend for sales by year and month by different product categories. We can simply copy the existing pivot table to a new worksheet and work from there.

Filters		Columns	
		Product Category	
Rows		Σ Values	
Years		Sum of Sales	
Order Date			

Sum of Sales	Column Labels					
Row Labels	Audio	Cameras and camcorders	Cell phones	Computers	Music, Movies and Audio Books	TV and Video
2018	182,424	6,750,412	2,196,698	6,635,135	426,434	2,728,048
Feb	6,504	466,840	149,413	466,647	32,341	167,206
Mar	8,254	486,074	187,716	561,056	42,948	162,191
Apr	24,332	433,028	150,856	424,247	44,795	179,451
May	7,940	597,172	210,988	816,645	32,167	258,762
Jun	29,136	679,810	281,364	789,193	32,827	239,643

We can add a PivotChart that is a line chart and stacked line chart to get an initial view of this data.



You can select which particular product category to project a trend line on if you wish.

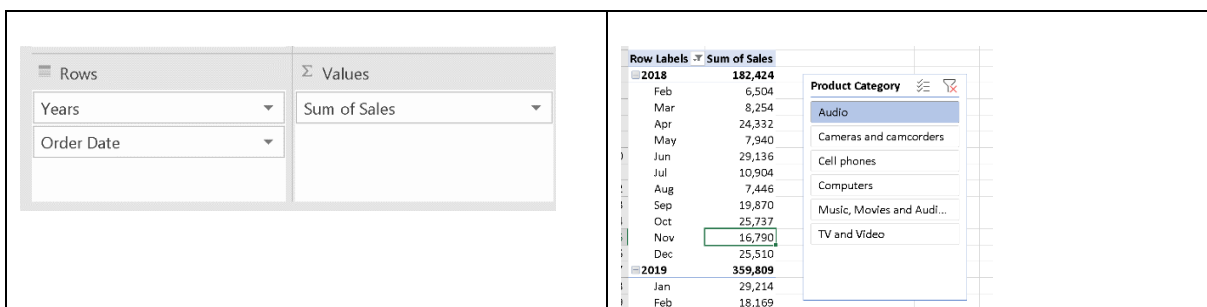


We can also change this to either an area graph and stacked area graph to get a better view of the contributions of each product category trending over time.





You could also take your Product Category out of the Columns and utilize it as a slicer instead, which then allows you to view the trends for each of these categories across the different chart types.



The next thing we will look is a seasonality graph, which we can create from another copy of the pivot table on a new worksheet. For seasonality, we typically want to take into account the years which have data for all the months, and filter out those that do not.



Row Labels	Sum of Sales
2019	17,741,637
Jan	2,081,908
Feb	1,252,298
Mar	1,507,751
Apr	1,103,451
May	1,408,131
Jun	1,657,980
Jul	1,248,653
Aug	1,593,694
Sep	1,340,433
Oct	1,300,579
Nov	1,511,809
Dec	1,734,950
2020	17,284,124
Jan	1,824,856
Feb	1,168,697
Mar	1,234,561
Apr	1,345,859
May	1,318,429
Jun	1,347,578
Jul	1,656,039
Aug	1,507,838
Sep	1,452,020
Oct	1,499,587
Nov	1,374,350
Dec	1,554,311
Grand Total	35,025,760

Next change custom grouping to Month. This will combine data from all the years concerned (2019, 2020).

Row Labels	Sum of Sales
2019	
Jan	
Feb	
Mar	
Apr	
May	
Jun	
Jul	
Aug	
Sep	
Oct	
Nov	
Dec	
2020	
Jan	
Feb	
Mar	
Apr	
May	
Jun	
Jul	
Aug	
Sep	
Oct	
Nov	
Dec	
Grand Total	

Grouping

Auto

☒ Starting at: 2/2/2018

☒ Ending at: 2/2/2021

By

Minutes

Hours

Days

Months

Quarters

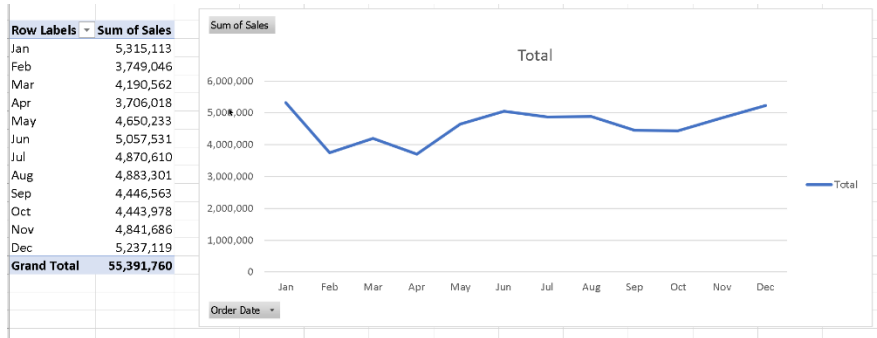
Years

Number of days: 1

OK Cancel

Row Labels	Sum of Sales
Jan	5,315,113
Feb	3,749,046
Mar	4,190,561
Apr	3,706,018
May	4,650,233
Jun	5,057,531
Jul	4,870,610
Aug	4,883,301
Sep	4,446,563
Oct	4,443,978
Nov	4,841,686
Dec	5,237,119
Grand Total	55,391,760

Then we can generate a line graph to demonstrate the Seasonality.



Seasonality is really useful to understand so that we can understand where resource / sales demand peaks in specific periods of a year and therefore can significantly help with resource allocation planning. Its important to make sure you have a complete set of data for the entire time duration you are interested to perform analysis on (for e.g. all the months of a year, rather than partial months) because that will skew the visualization of the graph.

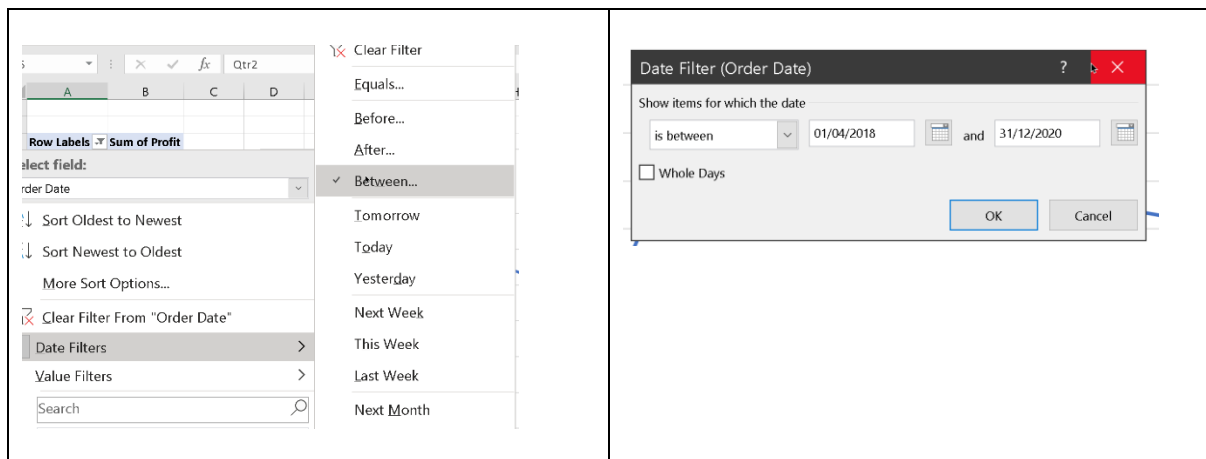
3.1 Practical exercise for Trend Analysis

Analytical activities to perform:

- Create a line graph displaying the Profit by Year and Quarter
- Add a polynomial trend line to the line graph
- Add a Slicer for Product Category
- Create an area graph displaying the Profit by Year and Quarter for the Product Category
- Add a Slicer for Country

File to use: File to use: SalesData-v5.xlsx

For first solution a), we can specifically target the quarters which are outliers in the dataset (suggesting that the data might be incomplete) and remove them using a Date filter with a Between.



Row Labels	Sum of Profit
2018	10,266,628
Qtr2	3,352,068
Qtr3	3,431,388
Qtr4	3,483,171
2019	11,378,957
Qtr1	3,172,056
Qtr2	2,541,105
Qtr3	2,731,591
Qtr4	2,934,205
2020	11,237,878
Qtr1	2,705,240
Qtr2	2,588,481
Qtr3	3,037,253
Qtr4	2,906,905
Grand Total	32,883,463